

* Basic Security Terminology

Computer security entails the methods used to ensure that a system is secure.

↳ Subjects such as authentication and access controls must be addressed in broad terms of computer security.

* The "CIA" of Security

The goal of computer security has been threefold: Confidentiality, integrity and availability.

Confidentiality:

It is to ensure that only those individuals who have the authority to view a piece of information may do so.

No unauthorized access.

Integrity:

It is related concepts that deals with the generation and modification of data.

↳ only authorized individuals should ever be able to create or change information.

Availability:

It is to ensure that the data, or the system itself, is available for use when the authorized user wants it.

$$\text{Protection} = \text{Prevention} + (\text{Detection} + \text{Response})$$

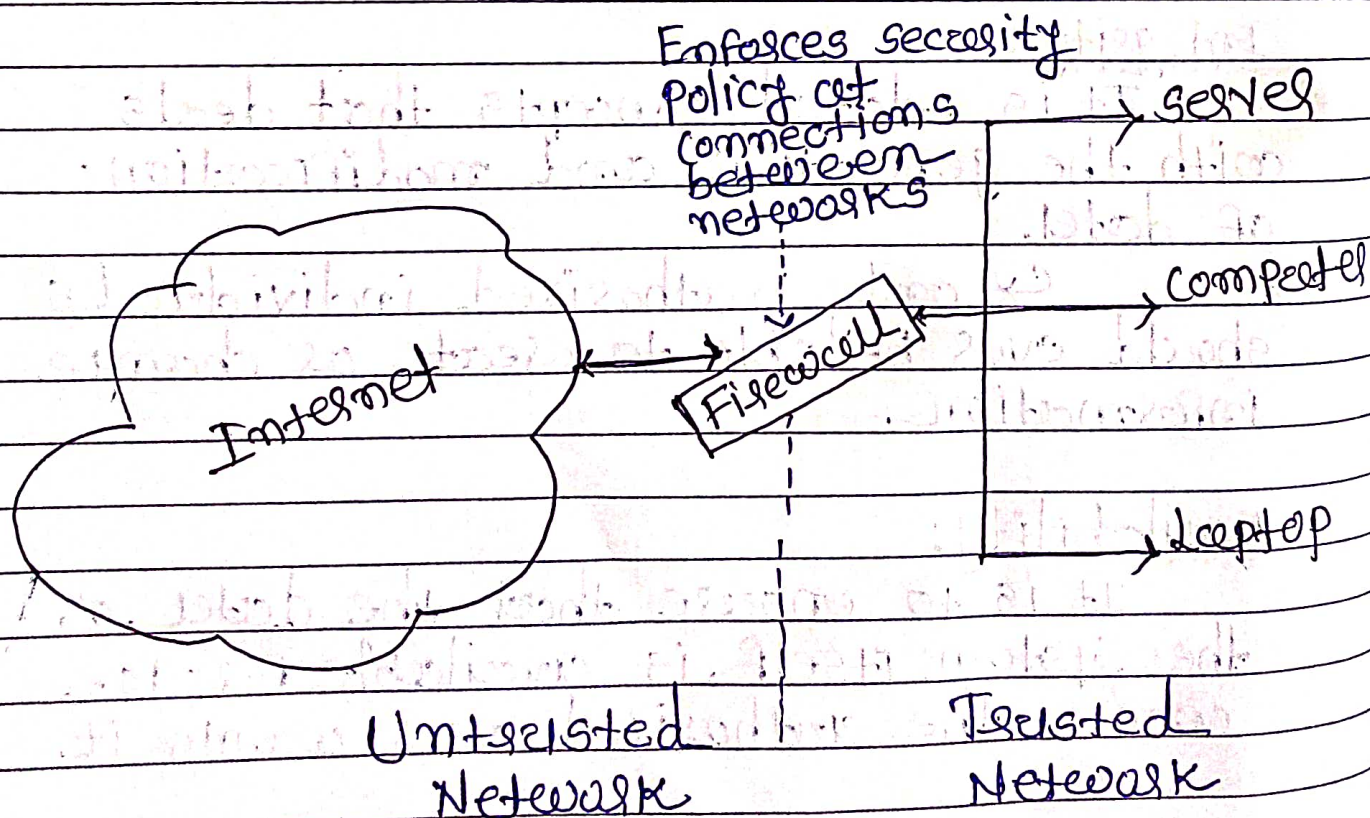
* Firewalls:

A Firewall is a network device - hardware, software, or a combination thereof - whose purpose is to enforce a security policy across its connections by allowing or denying traffic to pass into or out of the Network.

A Firewall is a lot like a gate guard at a security facility.

↳ The guard examines all the traffic trying to enter the facility.

A heart of a Firewall is the Firewall policy that it enforces.



* Types of Firewall:

Firewall enforces the established security policies. They can do this through a variety of mechanisms.

a. Basic Packet Filtering:-

Basic packet filtering looks at each packet entering or leaving the network and then either accepts the packet or

↳ rejects the packet based on user-defined rules. Each packet is examined separately.

b. Stateful Packet Filtering:-

Stateful packet filtering also looks at each packet,

↳ but it can examine the packet in its relation to other packets.

So stateful firewall keep track of network connections and can apply slightly different rule sets based on whether the packet is part of an established session or not.